



Deepfake face detection via multi-level discrete wavelet transform and vision transformer

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Abstract

Face tampering, involving the modification of facial images using advanced software and deepfake technology, poses a significant security threat. Traditional face manipulation detection methods often struggle with limited frequency information, hindering their effectiveness. To address this, we propose a novel face tampering detection framework that combines multi-level discrete wavelet transform (DWT) and vision transformer. By extracting multi-level frequency features using DWT and capturing global inconsistencies across local face areas through the vision transformer, our approach reveals discriminative details and frequency information. We evaluate our method on public datasets, achieving state-of-the-art accuracy of 99.86% and 99.92% on FaceForensics++ and Celeb-DF, respectively. These results demonstrate the efficacy of our framework in detecting deepfake face images, paving the way for more robust facial manipulation detection systems. The source code of our proposed algorithm and the related datasets are available at https://github.com/shawn-zhangxiang/MLFFE_ViT.

Keywords Deepfake face detection · Discrete wavelet transform · Vision transformer · Deep learning · Computer vision

1 Introduction

Image manipulation is an advanced technology that allows users to modify digital images using specialized software tools or mobile applications. This technology is applied to create new, visually stunning content by cropping, resizing, retouching, splicing, and other techniques. In most cases, image manipulation is used to beautify images, and people share their knowledge and expressions regarding the photo editing of visual arts. However, it becomes crucial when these techniques are used with evil intentions to make forged faces from existing authentic faces that are realistic and indistinguishable from pristine ones. It is challenging to recognize the tampered face images even with careful inspection.

Detecting the image's authenticity, especially the facial image, is crucial in criminal investigation, forensic case investigation, fraud insurance claims, fake journalism, and

forged digital images [1]. Therefore, different image tampering detection techniques are proposed to identify fake face images. In recent times, there have been significant dedicated endeavors to identify face tampering which have produced desirable outcomes in the area of forgery face detection. Various existing methods for detecting forged faces, many researchers have utilized discrete cosine transform (DCT) [2, 3] and discrete Fourier transform (DFT) [4] to incorporate contextual information from spatial and frequency domains. The discrete cosine transform is employed to convert the RGB image into the frequency domain in order to enhance the concealed forged trace in the high-frequency domain [3]. Jie et al. [5] introduce a multi-scale wavelet transformer-based method to detect tampered face images that utilize wavelet decomposition's multi-scale properties to enhance the aggregation of features across the frequency and spatial domains.

The use of multi-scale discrete wavelet transform (DWT) is highly effective in detecting deepfakes, as these manipulated images often incorporate subtle changes that are not readily identifiable through single-scale analysis. The multi-scale approach enables detection of these changes at multiple levels of resolution. However, most models solely concentrate on extracting only one level of frequency information extraction which limits the obtaining adequate frequency

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information. The multi-level frequency obtained by discrete wavelet transform (DWT) is more potent for getting enough frequency information.

The vision transformer [6] has utilized in face manipulation detection due to its remarkable ability to capture long-range dependencies and contextual relationships within images, which are essential for identifying subtle forgeries. Vision transformer, in contrast to conventional CNN, employ self-attention mechanisms to analyze the entire image as a sequence of patches. This approach allows them to more effectively capture global context and interactions between distant regions of the image. This comprehensive insight is especially advantageous for identifying face forgeries, as these manipulations often include intricate, spatially disseminated changes that localized analysis might uncover. In order to detect image tampering, various techniques involve the combination of discrete wavelet transform and vision transformers. In our framework, we utilize Haar DWT [7] to obtain multi-level frequency features and apply the vision transformer for capturing global inconsistencies across extracted local features. The integration of discrete wavelet transform (DWT) with vision transformer (ViTs) seeks to improve deepfake detection through the combination of their respective positive aspects. DWT effectively captures intricate frequency domain features, whereas ViTs excel in capturing long-range dependencies and providing a comprehensive global context. This integration utilizes the strengths of both approaches to enhance detection performance. The combination of multi-scale 2D-DWT and vision transformer output logits demonstrates strong performance in producing high-quality results.

The main highlighted points of the paper are given below:

- We propose a deepfake face detection approach combining the output of the multi-level discrete wavelet transform (DWT) and vision transformer for obtaining the adequate frequency features, distinguishable information and global forgery cues.
- To extract frequency features, we have applied a multi-level DWT module, which solves the existing one-level frequency extraction constraint, and the vision transformer obtains the global inconsistencies across the local face area.
- We have evaluated our proposed framework on the deepfake dataset and obtained higher AUC and F1-scores collated with the state-of-the-art model's results.

2 Related work

With the help of compelling image modification applications and tools, fake face detection becomes a hugely challenging

task. Image manipulation detection techniques can be categorized into two major categories according to the method employed for extracting features: traditional techniques and deep learning-based techniques.

2.1 Traditional techniques

In the traditional method, the handcrafted feature-based technique is used to capture statistical fingerprints and methods for analyzing specific patterns preserved in forged images [8–11]. Different types of handcrafted techniques are designed based on handcrafted features such as noise inconsistency (NOI) [12], color filter array (CFA) [13], error level analysis (ELA) [14], local bidirectional coherency error [15], and noise pattern [16]. Furthermore, these methods frequently exploit temporal and spatial inconsistencies in videos, looking for variations from frame to frame that could reveal manipulation. Detecting the artifacts applied to these handcrafted feature-based techniques became more challenging because of the evolution of generative adversarial network (GAN) models and different highly developed multimedia data compression methods. Also, the detection performance of these techniques is not good enough due to their robustness, high accuracy, and generalization. Deep learning-based frameworks have been applied that produce distinct outputs to solve these problems.

2.2 Deep learning techniques

Deep learning-based spatial feature extraction techniques are designed to improve the present method's performance. These techniques are mainly based on attention [17], consistency [18], and domain generalization techniques [19]. Wodajo et al. [20] designed a convolutional vision transformer (CViT) method to identify deepfake videos by combining the convolutional neural network and vision transformer. In this method, CNN is applied to extract learnable features, and the vision transformer is used to classify the results using the attention mechanism.

Chollet et al. [21] introduced a revolutionary neural computer vision architecture called the Xception framework. This framework replaces Inception modules with depthwise separable convolution. It shows revolutionary detection abilities on several publicly available deepfake datasets and has been used as a benchmark across computer vision tasks. The design of GocNet [22] aimed to detect AI-generated face forgery. In this model, the tensor preprocessing (TP) and manipulation trace attention (MTA) modules employ the image gradient operator to precisely find subtle traces. Zhao et al. [17] designed a multi-attentional deep fake detection framework for obtaining discriminative detailed information to detect tampered face images by enhancing the texture features from shallow layers and multiple attention maps. To

evaluate the cross-dataset, some researchers focus on the generalization of the detector. L. Li et al. [23] developed an effective, more generalized face forgery detector, Face X-ray, for detecting tampered face images. An efficient face forgery detection approach is proposed that utilizes the ViT backbone, initialized with ImageNet weights, and employs the low-rank adaptation method [24]. This framework reduces trainable parameters, minimizes computational resources, and further improves model generalization.

Frequency information is widely used in image forgery detection to obtain forgery traces for image enhancement, classification, and denoising. In [25], they designed a face forgery detection network using a swin-transformer to obtain intra-frame and intermodal discrepancies across the spatial and frequency domains. Qian et al. [2] introduced the F3-Net technique for tampered face detection, which employs the discrete cosine transform (DCT) to extract frequency feature. In WaveMix [26], the trainable convolution is used before multi-level 2D-DWT to obtain lossless transforms after each level of decomposition, image-suitable down-sampling, and token-mixing, which requires less GPU RAM and results in higher accuracy. Therefore, the transformer [27] has been used in computer vision and has significant advantages in modeling long-range dependencies. The computer vision field has effectively applied vision transformer [6] for segmentation [28], image classification [29], and object identification [30]. The efficient attention pyramid transformer (EAPT) [31] utilizes deformable attention, the encoder-decoder module, and the multi-dimensional continuous mixture descriptor for obtaining more comprehensive global attention information among patches and enhancing classification, object detection, and semantic segmentation performance.

An uncertainty-guided hierarchical frequency domain transformer (HFDT) technique has been introduced for obtaining low- and high-frequency information to provide degraded prior information and increase the image resolution [32]. Another method, Trans-FCA [33] is proposed to add global transformer features extracted by a transformer with local convolutional features. An F2Trans is a high-frequency fine-grained transformer designed for extracting inherent detailed features and high-frequency components to detect forged face images in spatial and frequency domains [34]. These methods consider a single level of frequency information, which is one of the limitations of the above techniques. Extraction of abundant frequency information requires more than one frequency utilization level, and multi-level frequency features contain more distinguishable details to identify forged images.

The combination of convolutional neural networks (CNNs) and vision transformers (ViTs) in face forgery detection exploits the advantages of both architectures. Convolutional neural networks (CNNs) efficiently capture local

features and spatial details, whereas ViTs excel at modeling long-range dependencies and global context [35]. By integrating these techniques, models can more effectively identify subtle anomalies in manipulated images. CNN-based architectures like EfficientNet [36] and Xception are utilized for feature extraction, and the resulting output is then passed as a sequence to a transformer [37–39]. In Vixnet [39], Xception is utilized for extracting global spatial features, while the vision transformer is employed for identifying artifacts that manifest in various facial regions. This hybrid methodology improves the accuracy and robustness of detection systems, rendering them more resistant to sophisticated deepfake techniques and making a significant contribution to advancements in digital forensics.

In our proposed method, we use discrete wavelet transform to obtain multi-level frequency features, and the use of vision transformer in our network aims to leverage the presence of leftover imperceptible artifacts. We integrate the output logits from these into a feed-forward network to classify face manipulation detection.

3 Proposed method

Given an input image, we aim to detect the tampered face image. Our proposed model, demonstrated in Fig. 1, is a DWT and vision transformer-based framework consisting of two main modules multi-level frequency feature extraction (ML-FFE) and vision transformer (ViT) to exploit different artifacts, especially for detecting deepfake face images. In the ML-FFE module, we employed a multi-level discrete wavelet transform (DWT) to extract more frequency features and acquire abundant frequency information. In addition, we employed the vision transformer module to identify global inconsistencies among the extracted local features. A detailed description of each module is given below:

3.1 Multi-level frequency feature extraction

The first module of our proposed approach is multi-level frequency feature extraction (ML-FFE) module, which is the focal module for detecting tampered face images. Discrete wavelet transform (DWT) is widely used in computer vision tasks. Using DWT, we can represent multi-scale frequency features. It has also been used to achieve remarkable performance in image denoising [40], restoration [41], image depth prediction [42], compression, and fusion [43]. A 2D-DWT splits an image into four frequencies containing one approximation LL low frequency and three detail, LH, HL, and HH high-frequency sub-bands [44]. In this module, we extract the multi-level frequency features from the face image by applying the Haar wavelet [7]. It has more comprehensive

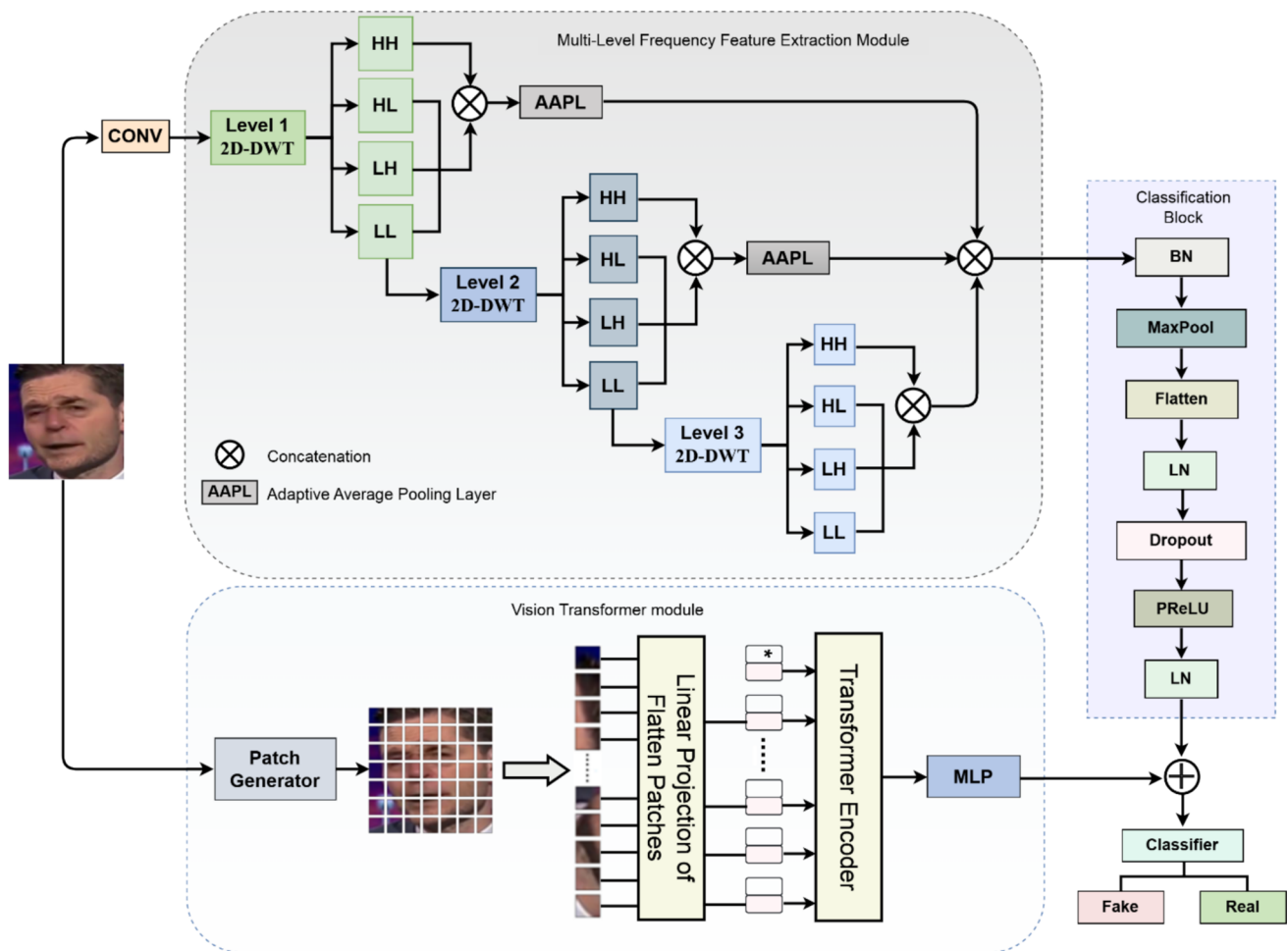


Fig. 1 Deepfake face image detection framework using multi-level 2D-DWT and vision transformer

details that can be utilized to differentiate forged and authentic images effectively. The multi-level frequency features can also fix one-level frequency limitations. Initially, we pass the input image $X \in \mathbb{R}^{H \times W \times 3}$ through a convolutional layer to increase the number of channels for embedding dimension C and reduce C by a factor of four for obtaining the same number of channels in the concatenated result after each level of 2D-DWT decomposition. Then, the 2D-discrete wavelet transform divides a grayscale image into three high-frequency (LH, HL, and HH) and one low-frequency (LL) sub-band. LL represents low frequency, and the other three components, LH, HL, and HH, incorporate high-frequency information that can detect tiny distortions and forgery traces of tampered face images. Again, the approximation sub-band (LL) is decomposed applying a repeated 2D-discrete wavelet transform for achieving next level transform and so on. We perform a channel-wise concatenation (cat) operation to aggregate the LL, LH, HL, and HH features. The following formula represents the overall procedure:

$$\tilde{X}_l = \text{cat} \left(X_{LL}^l, X_{LH}^l, X_{HL}^l, X_{HH}^l \right),$$

$$\text{where } \tilde{X}_l \in \mathbb{R}^{\frac{H}{2^l} \times \frac{W}{2^l} \times 4C/4}, l \in \{1 \dots L\} \quad (1)$$

$$\tilde{X}_1 = A(\tilde{X}_1) \quad (2)$$

$$\tilde{X}_2 = A(\tilde{X}_2) \quad (3)$$

$$\tilde{X} = \text{cat}(\tilde{X}_1, \tilde{X}_2, \tilde{X}_3) \quad (4)$$

where $\tilde{X}_1$, $\tilde{X}_2$, and $\tilde{X}_3$ represent other outputs of levels 1, 2, and 3, respectively, A indicates the adaptive average pooling layer, and $\tilde{X}$ is the final concatenation output.

We decompose 2D-DWT up to level 3 for extracting multi-level frequency information. Using more levels of 2D-DWT provides a more receptive field but also increases noise in the model, which reduces the model's effectiveness [45].

To match with the level 3 decomposition feature map, we apply the adaptive average pooling layer A in the first and second levels, respectively. Finally, the concatenated outputs from each level of DWT are concatenated together, and the obtained feature map has passed through the classification block. The classification block is shown in Fig. 1, and it consists of a batch normalization (BN) layer, max-pool, flatten, dropout, and parametric rectified linear unit (PReLU) activation function and linear layer (LN). We utilize PReLU activation function which provides more flexibility in learning and BN for stabilizing activations and gradients, allowing for faster training and potentially better performance.

3.2 Vision transformer module

In the second module, we use the vision transformer (ViT) to identify global discrepancies among the patches generated by the patch generator. There are multiple transformer blocks in the vision transformer, which is the fundamental component, and each transformer block comprises several multi-head self-attention blocks along with feed-forward neural networks and layer normalization. The extraction of global discrepancies among the patches is subsequently achieved by employing multi-headed self-attention. A vision transformer encoder block architecture and multi-head self-attention are shown in Fig. 2. Initially, the input image of dimension $(N \times N)$ is passed through the patch generator that splits the image into smaller M patches of dimension size $(P \times P)$, where $M = (N \times N)/P^2$. The achieved patches from patch generator are embedded into linear patch embeddings $X \in \mathbb{R}^{M \times d}$, where X represents the input patch embeddings and d for the embedding dimension. Next, positional encodings are added to incorporate positional information into the patch embeddings. The result of the embedding is passed to the transformer encoder as input.

Multi-head self-attention takes the *Query*(Q), *Key*(K), and *Value*(V) as input matrices by packing the embeddings into a matrix X and multiplying it by the weight matrices $W_Q \in \mathbb{R}^{d \times d_Q}$, $W_K \in \mathbb{R}^{d \times d_K}$, and $W_V \in \mathbb{R}^{d \times d_V}$ which is shown in Eq. (1).

$$Q = XW_Q, K = XW_K, V = XW_V \quad (5)$$

The Key(K) and Query(Q) generate the attention output. The dot product of the Q and K transpose matrix is calculated and divided by the scale factor. Then, the softmax function is applied to this score matrix and multiplied by the V matrix to determine the attention score. The attention score is calculated using the following formula:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (6)$$

where d_k represents embedding dimension.

The result of multi-head self-attention is passed to MLP through a position-wise feed-forward neural network (FFNN). In this module, a pre-trained ViT-B_32 transformer model with a larger patch size of 32×32 is utilized to avoid the need to train ViT from scratch which makes it more feasible in terms of both computational cost and time. The pre-trained ViT model effectively captures global contextual information and enhances the model's ability to detect subtle and global inconsistencies in manipulated images. The obtained output logit from MLP is fused with multi-level frequency feature extraction module's output logit generated by classification module. Finally, the fused logit is sent to the classifier for obtaining the manipulation identification results. The combination of these two modules produces superior classification performance than applying them individually.

4 Experiments and results analysis

In this section, we describe the dataset and details of our implementation, taking into account the two most popular and publicly available deepfake video datasets, Celeb-DF [46] and FaceForensics++ (denoted as FF++) [47], as well as one image dataset, deepfake image dataset (DFID) [48]. Next, we describe how these datasets are set up for training, testing, and validation process. Finally, we assess the proposed method's performance by comparing its outcomes with the present modern methods.

4.1 Dataset

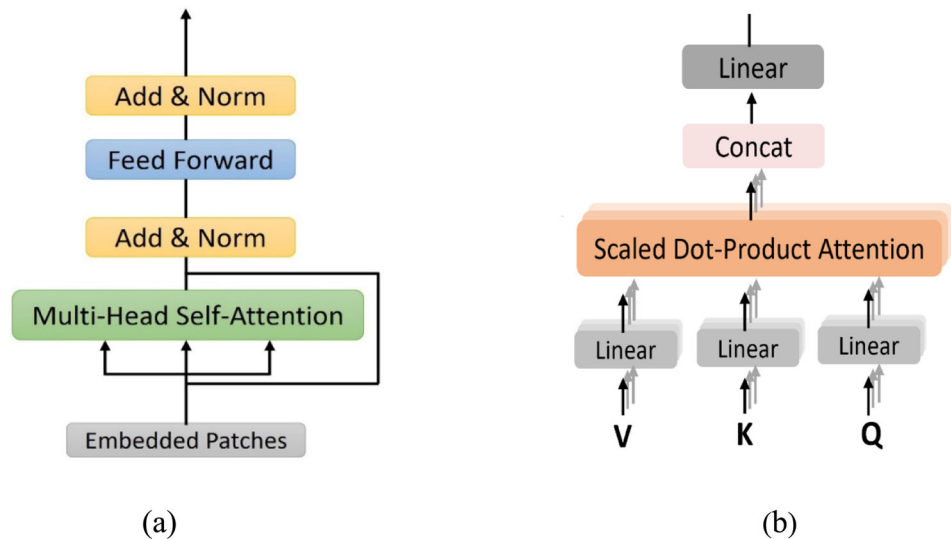
4.1.1 Celeb-DF

The Celeb-DF dataset is an extensive compilation specifically created for the purpose of conducting research on deepfake detection. The content consists of videos in high-resolution featuring realistic facial manipulations, effectively reducing any visual distortions. The dataset showcases a broad array of celebrities, providing varied examples that present a challenge to detection algorithms. It comprises 5639 fake and 590 real high-resolution videos with high visual quality. Each video lasts around 13 seconds and has a frame rate of 30 frames per second. Li et al. [46] gathered authentic videos from YouTube that were publicly available, in contrast to 59 celebrities with various distribution methods.

4.1.2 FaceForensics++

The FaceForensics++ dataset is a critical resource when conducting research on the detection of facial manipulation. It

Fig. 2 **a** Transformer encoder block architecture and **b** multi-head self-attention block



acts as a benchmark for developing robust detection techniques, providing a significant contribution to the progress of digital forensics and media integrity. This dataset contains 1000 authentic videos collected from YouTube videos provided by Rossler et al. [47]. Next, each video was manipulated using four automated deepfake methods: Deepfakes, Face2Face, FaceSwap, and NeutralTexture. Finally, this dataset consists of 4000 fake videos and 1000 real videos. We concentrate on the deepfake method, which uses the face swapping manipulation technique to forge faces. The author prepared two different quality videos with high (HQ) and low (LQ) compression levels, and we consider both the HQ and LQ compression level for our implementation.

4.1.3 DFID

The deepfake image dataset (DFID)[48] is a compilation of manipulated pictures applied for the purpose of studying and developing methods to detect deepfake content. It offers a wide variety of examples, which assist in the training and assessment of machine learning models. The dataset comprises a total of 19,809 images, with 8,000 of them being forged images and the remaining 11,809 being real images. We utilize a total of 80% images for training purposes, 10% images for validation, and 10% images for testing.

4.2 Pre-preprocessing

We have opted to apply a preprocessing step to prepare the videos of the Celeb-DF and FF++ datasets for facial recognition. We extract multiple frames from each video and apply an MTCNN algorithm [49] to determine which facial part is apparent in each frame. The MTCNN algorithm is highly efficient in detecting faces, aligning them and accurately

localizing facial landmarks in both images and videos. The frame interval sampling approach is applied to select frames during the video processing stage for ensuring both temporal diversity and high-quality face detections. Subsequently, the extracted face images are saved into 256×256 dimension, ensuring a 10% padding around the detected face area. Later, we resize images to a fixed dimension of 224×224 , maintaining the aspect ratio throughout the model's training, validation, and testing stages. Fig. 3 shows a few samples fake and authentic face images.

We extracted more than 100k face images for each category of real or fake videos and manually removed images that contained incomplete or partial faces, hands, actors' clothing, and other items in the video background. We ultimately choose 50k, 10k, and 10k images for each category for training, validation, and testing process, respectively. To evaluate the performance on the FF++(LQ) compression level, we simply selected 20k out of 25k images extracted following the above procedure and use 70% images for training, 20% for validation, and 10% for testing procedures.

4.3 Experiment setup

We choose 32 as a batch size and 0.0001 for learning rate in order to train the model. A low learning rate allows the model to make gradual adjustments, effectively reducing loss and ensuring the model's convergence toward the best solution. Because the classification's output is binary, we opted to deploy binary cross entropy loss function and Adam is the optimizer. Our proposed model is implemented using the PyTorch framework and utilizes the NVIDIA GeForce RTX 3090 Ti GPU for all experiments.

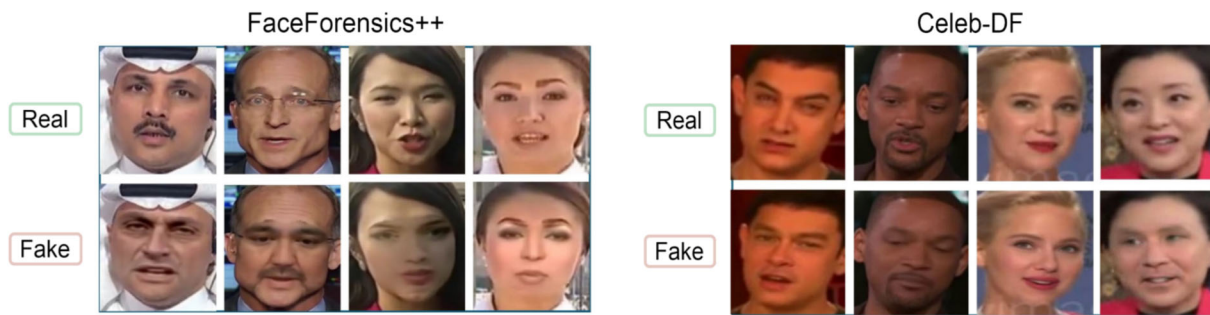


Fig. 3 The result of preprocessing real and fake images from Celeb-DF and FF++ (HQ) datasets

4.4 Evaluation metrics

We show the method's performance in the experiment using testing accuracy, F1-score, and AUC evaluation metrics.

4.4.1 Accuracy

Accuracy (ACC) is the widely used evaluation metric that measures the proportion of the correct predictions and the sum of all predictions. Using the following formula, we determine the accuracy by counting the instances of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN):

$$ACC = \frac{TP + TN}{TP + TN + FP + FN} \quad (7)$$

A high ACC score indicates good predictions, and a low score represents poor predictions the model makes.

4.4.2 AUC

The AUC score is a useful metric for comparing and assessing classifier performance. It represents the classification model's performance and indicates the tradeoff between the actual positive rate and the false positive rate. Mathematically, true positive rate (TPR) and true positive rate (FPR) can be written as follows:

$$TPR = \frac{TP}{TP + FN} \quad (8)$$

$$FPR = \frac{FP}{FP + TN} \quad (9)$$

4.4.3 F1-score

F1-score is an evaluation metric that integrates *Precision* (P) and *Recall* (R) and is used to represent the model's performance in binary classification problems in machine learning and statistics. A higher score signifies

better performance on the F1-score scale, which ranges from 0 to 1. The following formula is employed to compute the F1-score:

$$P = \frac{TP}{TP + FP}, R = \frac{TP}{TP + FN}$$

$$F1 = \frac{2 \times P \times R}{P + R} \quad (10)$$

4.5 Performance analysis

This section shows the model's achieved results for the FaceForensics++, Celeb-DF, and DFID datasets. We apply the same training, validation, testing procedure, and parameter configurations to produce unbiased results from experiments and comparisons with the latest models. We also apply the same evaluation metrics used for datasets and other present methods here.

Table 1 presents our model's results on the FaceForensics++ dataset. We achieved 99.86% accuracy, 0.998 F1-score, and 0.9995 AUC score in the high-quality (HQ) compression level. In the case of low-quality compression levels (LQ), we get 97.00% accuracy, which is the best performance compared to other methods. Our approach demonstrates superior outcomes in comparison with convolutional neural network frameworks such as EfficientNet and Xception. CNN, when combined with a vision transformer such as Vixnet, demonstrated superior performance compared to simply employing a vision transformer alone. Our proposed method, referred to as **(ML-FFE+ViT)Ours**, outperformed all other methods indicating that the network effectively extracts useful frequency features using multi-level DWT, and the vision transformer efficiently captures global inconsistencies among local face region. The table clearly shows that the combination of the output logit of multi-level frequency feature extraction (ML-FFE) and ViT module achieved a higher accuracy than the individual classification result.

Table 1 The methods' results employing the FaceForensics++ dataset

Method	ACC (%)		Precision (%)	Recall (%)	F1-score	AUC
	HQ	LQ				
ViT [6]	88.56	80.89	89.61	87.77	0.886	0.9524
EfficientNetB0 [36]	86.37	69.04	88.89	84.83	0.868	0.9369
Xception [21]	99.75	88.91	99.85	99.50	0.996	0.9993
CNN-ViT [20]	93.72	78.04	88.68	98.63	0.933	0.9961
Vixnet [39]	99.71	96.48	99.72	99.74	0.997	0.9930
GocNet [22]	98.98	88.82	98.85	99.12	0.989	0.9950
WaveMix [26]	99.48	96.77	99.06	99.91	0.996	0.9962
(ML-FFE) Ours	99.61	96.64	99.61	99.63	0.996	0.9963
(ML-FFE + ViT) Ours	99.86	97.00	99.82	99.91	0.998	0.9995

Table 2 The methods' results employing the Celeb-DF dataset

	ACC (%)	Precision (%)	Recall (%)	F1-score	AUC
ViT [6]	86.14	85.96	86.39	0.862	0.9372
EfficientNetB0 [36]	80.55	80.77	80.15	0.805	0.8762
Xception [21]	99.80	99.85	99.78	0.998	0.9994
CNN-ViT [20]	98.64	99.86	97.41	0.996	0.9980
Vixnet [39]	99.75	99.86	99.61	0.997	0.9987
GocNet [22]	98.91	99.17	98.66	0.989	0.9992
WaveMix [26]	99.85	99.86	99.50	0.996	0.9994
(ML-FFE) Ours	99.86	99.84	99.88	0.998	0.9993
(ML-FFE + ViT) Ours	99.92	99.89	99.96	0.999	0.9996

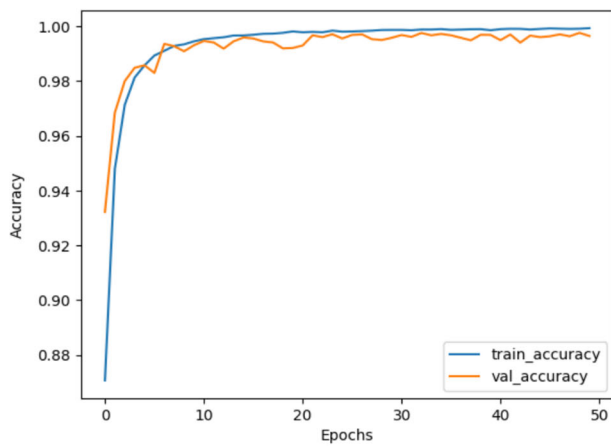
Table 3 The methods' results employing the DFID dataset

Method	ACC (%)	Precision (%)	Recall (%)	F1-score	AUC
ViT [6]	96.10	93.44	96.16	0.947	0.9879
EfficientNetB0 [36]	90.88	87.02	90.92	0.889	0.9651
Xception [21]	99.16	99.25	98.26	0.987	0.9992
CNN-ViT [20]	98.94	98.99	98.49	0.987	0.9988
Vixnet [39]	98.78	98.56	98.01	0.982	0.9987
GocNet [22]	97.30	98.11	95.58	0.968	0.9977
WaveMix [26]	98.94	98.87	98.62	0.988	0.9990
(ML-FFE) Ours	98.72	98.78	98.56	0.986	0.9920
(ML-FFE + ViT) Ours	99.26	98.18	99.54	0.993	0.9994

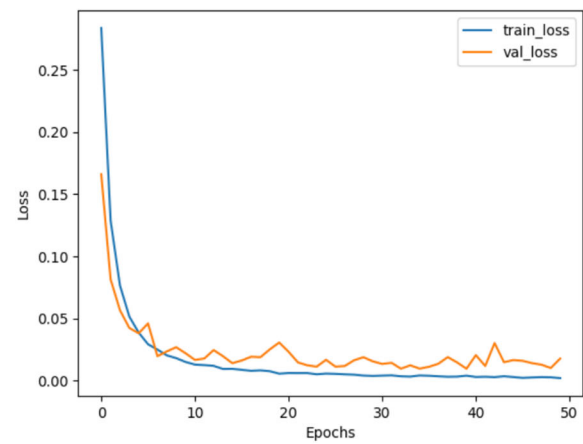
The performance outcomes of our suggested model on the Celeb-DF deepfake dataset are shown in Table 2. The table shows that our model also has achieved the best performance compared to existing methods. Our model testing accuracy is 99.92%, F1-score is 0.999, and AUC is 0.9996. Here also, the WaveMix method has given good results compared to other methods. Its performance is almost similar to our proposed model. WaveMix also applies multi-level frequency decomposition for image classification. Like the FaceForensics++ dataset, the utilization of convolutional neural networks (CNN) and vision transformers (ViT) in

models achieves superior performance compared to using simply CNN or ViT alone.

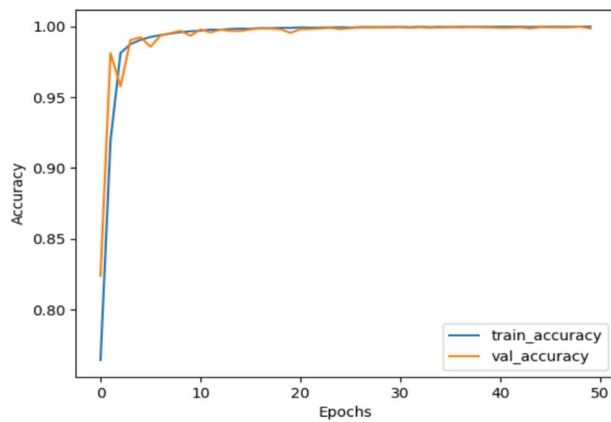
The DFID dataset trained the model to high performance, outperforming other methods with accuracy of 99.26%, F1-score of 0.993, and AUC score of 0.9994. Table 3 shows the proposed methods' results on DFID dataset. Here, compared to FF++ and Celeb-DF dataset, ViT and the EfficientNet model performed better. However, the remaining results of the model are nearly identical to those of the FF++ and Celeb-DF datasets. The performance on the DFID dataset is inferior to that of the FaceForensics++ and Celeb-DF datasets. The



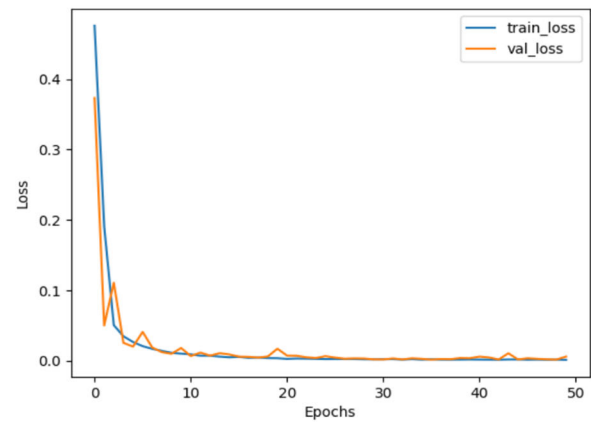
(a) Accuracy curve on FaceForensics++ dataset



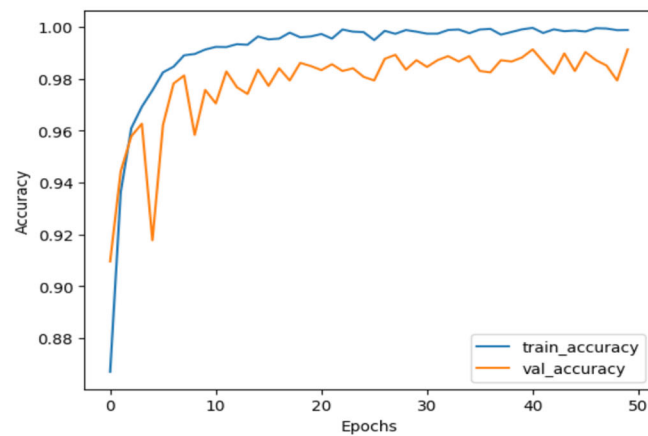
(b) Loss curve on FaceForensics++ dataset



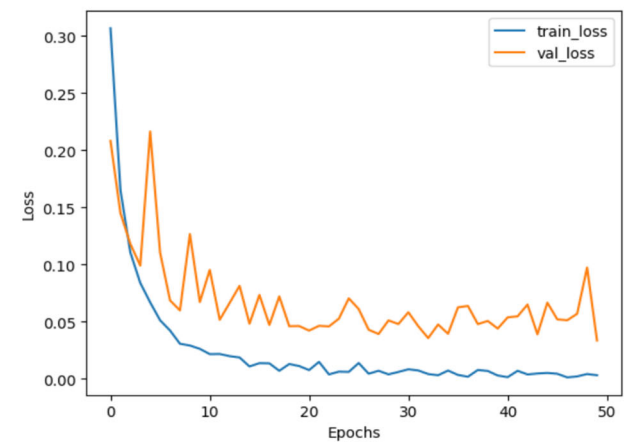
(c) Accuracy curve on Celeb-DF dataset



(d) Loss curve on Celeb-DF dataset



(e) Accuracy curve on DFID dataset



(f) Loss curve on DFID dataset

Fig. 4 **a** Accuracy curve on FaceForensics ++ dataset, **b** Loss curve on FaceForensics ++ Dataset, **c** Accuracy curve on Celeb-DF dataset, **d** Loss curve on Celeb-DF dataset, **e** Accuracy curve on DFID dataset, **f** Loss curve on DFID dataset

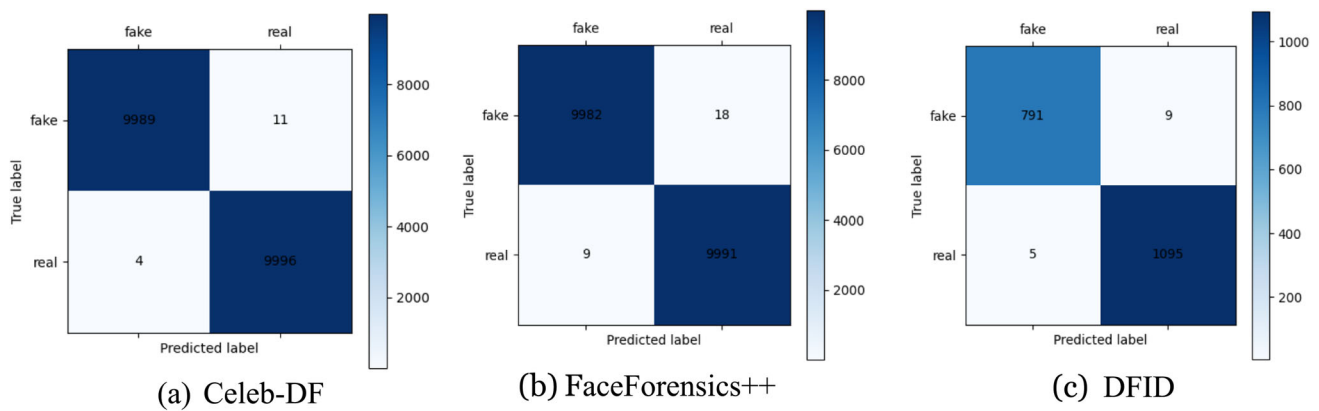


Fig. 5 Confusion matrices on the **a** Celeb-DF, **b** FaceForensics + + (HQ), and **c** DFID datasets

Table 4 Results of the 2D-DWT with various frequency level components (FF + +)

Method	Frequency Components with Level				ACC	Precision	Recall	F1-Score	AUC	Params
2D-DWT		LH ₁	HL ₁	HH ₁	87.25	87.78	86.55	0.8716	0.9460	13.50 M
	LL ₁	LH ₁	HL ₁	HH ₁	88.98	92.95	84.35	0.8844	0.9592	13.83 M
	LL ₂	LH ₂	HL ₂	HH ₂	89.43	86.09	94.05	0.8989	0.9661	15.14 M
	LL ₃	LH ₃	HL ₃	HH ₃	91.45	91.62	91.25	0.9143	0.9747	16.46 M
	LL ₄	LH ₄	HL ₄	HH ₄	92.85	96.22	89.20	0.9258	0.9826	17.77 M

suboptimal performance on DFID can be partially ascribed to its reduced dataset size in comparison with FF++ and Celeb-DF. This restricts data diversity and presents barriers to obtaining robust and generalizable features.

In Fig. 4, we present training and validation learning performance of our proposed model on FaceForensics + + , Celeb-DF, and DFID dataset. The validation accuracy and loss remain very close to the training accuracy and loss curves, respectively, with minimum overfitting. The obtained curves indicate that our model is properly trained and achieving high accuracy.

A confusion matrix is an essential tool for calculating the performance of classification models. In Fig. 5, we show the corresponding confusion matrix obtained by evaluating our model on Celeb-DF, FF + + (HQ), and DFID datasets.

The above confusion matrix result provides our model's excellent prediction accuracy compared to the ground truth levels. Our model can effectively distinguish between pristine and forged faces. As a result, the model provides a deficient number of false positives and false negatives.

5 Ablation study

This study looks into how different frequency sub-bands on feature extraction using the 2D-discrete wavelet transform (DWT) affect each other applying FF++ dataset. The process entails breaking down an image into sub-bands of low-frequency and high-frequency components, which yield approximate and detailed sub-bands. We can determine the contribution and significance of each frequency component by observing changes in performance metrics. Low-frequency components maintain image structure and smooth changes, while high-frequency components capture edges and intricate details for precise feature extraction.

In Table 4, we show the impact of 2D-DWT with different frequency decomposition level components. In this table, LL indicates the low frequency, and HL, LH, and HH represent the high-frequency components, where subscripts 1, 2, 3, and 4 represent frequency decomposition levels. From this analysis study, we observe that increasing the frequency decomposition levels increases the accuracy and AUC score; combining both low-frequency and high-frequency components produces better performance compared to solely utilizing high-frequency components. However, it also increases the number of parameters, which increases the calculation cost as well as more prone to overfitting. Comparing level

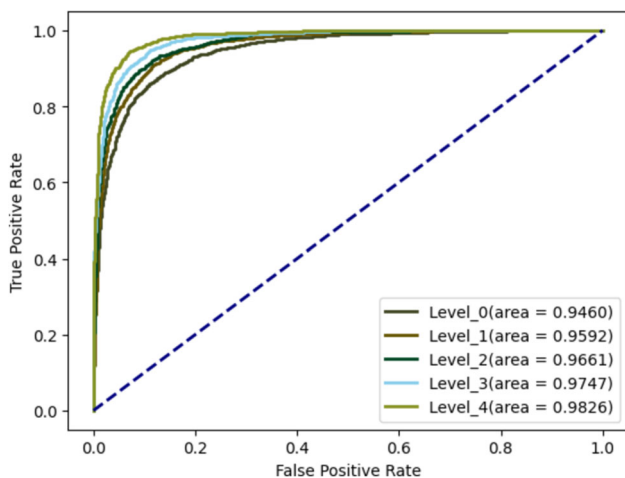


Fig. 6 ROC curves on multi-level frequency decomposition of 2D-DWT

1 and level 4, the total number of parameters is 13.83 million and 17.77 million, respectively. In our proposed method, we have used frequency decomposition level 3 for frequency feature extraction, which has more balanced precision and recall accuracy compared to other levels.

Fig. 6 presents the detailed ROC curves for the multi-level frequency decomposition of 2D-DWT. The figure clearly shows that as the frequency decomposition levels increase, the ROC curve position moves closer to the top left corner of the plot indicating superior performance.

6 Conclusion

This paper proposes a deep fake face detection approach combining the vision transformer and discrete wavelet transform. The multi-level frequency features are extracted using 2D-DWT, and the vision transformer is utilized to gather overall inconsistencies across the local features. Our proposed framework can achieve a state-of-the-art result on publicly available FaceForensics++ and Celeb-DF datasets, obtaining the AUC scores of 0.9995 and 0.9996, respectively. The classification accuracy, AUC, and F1-score show our method's best performance compared to other deepfake face detection methods. Our current work focuses only on deepfake face image detection, which limits our framework. In future work, we will explore the potential to extend our work to more complex tasks, such as deepfake detection in both images and videos.

Author contributions Main Uddin contributed to the conception of the proposed framework, performed the experiment, and prepared the manuscript; Zhangjie Fu made a valuable contribution by performing the analysis, engaging in constructive discussions, and reviewing the

manuscript; Zhang Xiang conducted an experiment analysis and edited the manuscript with constructive discussions.

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Data availability No datasets were generated or analyzed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

Ethical approval This declaration is not applicable.

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